MALMÖ UNIVERSITY

Introduction of smartphones in Farmer Field Schools

Digitalisation and communication practices in rural Mozambique

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Communication for Development
One-year master
15 Credits
2026
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Abstract

The digitalisation of agricultural development programmes is often framed as a solution to improve access to information, enhance communication in rural areas and increase productivity. In Mozambique, the Food and Agriculture Organization of the United Nations has recently introduced a digitalisation component into Farmer Field Schools (FFS) initiative, providing smartphones within a participatory learning approach grounded on face-to-face interaction and collective experimentation. This thesis examines how FFS participants, more specifically facilitators, interpret and negotiate the introduction of the smartphones within FFS groups in Nampula province. The study draws on information collected during three weeks of fieldwork conducted in Nampula in 2026, which combined focus groups, semi-structured interviews and participant observation. The analysis is informed by the core principles of Communication for Development, engages with debates on Information and Communication Technologies for Development and socio-technical perspectives from Information Systems. The findings underscore mismatches between the design of this digitalisation initiative and the realities faced by smartphone users. Smartphones were introduced with limited training, and the training provided shaped how the devices are used to communicate. Facilitators neither adopted nor rejected the smartphones as intended; instead, they adapted their use to existing communication practices and local needs. The study also identified multiple limitations that affect different forms of digital access and can contribute for patterns of digital inequality. The findings suggest that the adoption of new technologies by rural communities is not determined solely by their physical availability. It is a process shaped by the social, economic and institutional contexts in which the technology is introduced. Further research is needed to understand the long-term implications of the digitalisation of FFS in Mozambique.

Keywords: Communication for Development; ICT4D; Farmer Field Schools; Communication; Digitalisation; Smartphone; Mozambique; Agriculture.

Acknowledgements

I would like to express my deepest gratitude to my supervisor, Prof. Linda Austin, for her helpful guidance and support through this research journey. An important part of the process was our discussions that enriched my understanding of Communication for Development and encouraged me to explore the subject from different angles.

I would also like to express my sincere gratitude to all those who supported me during my fieldwork in Mozambique. I would like to thank the FAO team in Nampula, whose collaboration was instrumental in making this research possible. Their willingness to share their knowledge greatly enriched this study. I am equally grateful to Helpo, a Portuguese NGO that hosted me during my three weeks in Nampula. Their kindness made my stay smooth and allowed me to fully immerse myself in the local context. Without the support of these organizations and the remarkable people behind them, this project would not have been possible.

I also extend my appreciation to my colleagues and friends working in the field of development, whose support accompanied me throughout this journey. Many of them generously took the time to discuss ideas with me and provide feedback. I am especially grateful to Aurora Maggiotto, Joana Reis and Lorenzo Gentile for their encouragement during the research process.

I am thankful to all my friends and family, who stood by me throughout this experience. They have been a source of strength and reassurance at every stage of the process.

Finally, I dedicate this thesis to my mother, who raised me to be curious and open minded. She taught me that empathy should always be the core value of professional practice and that the most meaningful work is that which serves a purpose. Her unconditional support shaped both the person and the professional I strive to be.

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List of abbreviations

ADM	Agro-Negócio para o Desenvolvimento de Moçambique Lda.
ComDev	Communication for Development
CSA	Climate-Smart Agriculture
DSC	Development Support Communication Branch
EDF	European Development Fund
EMC	Escola na Machamba do Camponês
ERI	Enabling Rural Innovation
EU	European Union
FAO	Food and Agriculture Organization of the United Nations
FBS	Farmer Business School(s)
FFS	Farmer Field School(s)
FRELIMO	Front for the Liberation of Mozambique
GNI	Gross National Income
ICT	Information and Communication Technologies
ICT4D	Information and Communication Technologies for Development
ICT4Ag	Information and Communication Technologies for Agriculture
IFAD	International Fund for Agricultural Development
IMF	International Monetary Fund
INE	Instituto Nacional de Estatística
IPFRN	Instituto Politécnico Familiar Rural de Natete
IPM	Integrated Pest Management
IS	Information Systems
ITU	International Telecommunication Union
LMICs	Low- and Middle-Income Countries
UN	United Nations
UM DESA	United Nations Department of Economic and Social Affairs

1. Introduction

For more than half a billion smallholder farmers worldwide (FAO, 2025c, p. xvii), their livelihoods depend as much on access to information as they do on soil and rainfall. An estimated 500 million farms of less than two hectares are crucial for local food security, producing nearly 60 per cent of food in low and middle-income countries (LMICs) (FAO, 2025c, p. xiv). These farms are predominantly managed by some of the world's poorest households, 80 per cent of the world's poorest live in rural areas and rely heavily on agriculture (FAO, 2025d). Smallholder farmers in Sub-Saharan Africa adopt productivity-enhancing agricultural technologies at persistently low rates, they are held back by interrelated structural barriers, including limited access to credit, insurance, extension services and efficient input and output markets (Kumar & Doan, 2025, pp. 9-10). These constraints are intensified further by persistent pressure to align local farming practices with globalised and market-driven models of agricultural production (Lawrence & Smith, 2020). Such pressures are characteristic of the broader neoliberal restructuring of agrifood systems in the Global South, whereby market integration and commercial competitiveness are privileged, this often sits uneasily alongside smallholders' own practices and priorities (McMichael, 2009).

Mozambique illustrates this pattern closely. Agriculture remains the backbone of rural livelihoods in the country, where more than 70 per cent of households heavily rely on farming as a source of income (Bratley & Meyer-Cirkel, 2025, p. 3). The sector is characterized by persistently low productivity and high vulnerability to shocks, that constrain sustainable improvements in rural welfare (World Bank, 2026a, p. 4). It is within this context that international development actors, notably the Food and Agriculture Organization of the United Nations (FAO), have long implemented initiatives in the country aimed at improving smallholder farmers productivity and resilience. Among these efforts is the Farmer Field Schools (FFS) approach, locally known as Escola na Machamba do Camponês (EMC).

FFS is a people-centred learning approach where farmers test new agricultural knowledge and share experiences collectively, it aims to further develop farmers' analytical skills and strengthen local capacities to make decisions (FAO, 2016, p. 2). In line with the broader trend of integrating digital components into development projects, FFS in Mozambique has entered a phase of digital transformation part of another programme. In 2024, FAO distributed smartphones to FFS facilitators, and in April 2025, FAO and Mozambique's Ministry of

Communications and Digital Transformation launched a new initiative to digitalise FFS across the country. It aims to provide farmers with climate and market information, remote technical assistance and digital learning tools and it targets approximately 3,000 beneficiaries across 13 districts in Nampula, Zambézia and Manica [1]. As stated in FAO's press release, "The Digitalization of the FFS Project marks a significant step towards modernizing agricultural education in Mozambique, empowering farmers with technology to enhance productivity, market access, and climate resilience" (FAO, 2025a).

However, the introduction of new Information and Communication Technologies (ICTs) into rural development initiatives raises concerns that have long been highlighted in the Information and Communication Technologies for Development (ICT4D) literature, particularly regarding technological determinism, unequal access and the possible mismatch between technological interventions and local realities (Heeks, 2002; Toyama, 2015). These critiques are not new, but its application to the specific agricultural realities of Mozambique remains underexplored. Integrating smartphones into an initiative traditionally grounded in face-to-face learning requires careful consideration of local communication practices, power dynamics and digital trust. Moreover, structural barriers, including limited digital literacy, weak connectivity and inadequate infrastructure, pose significant constraints on the effective use of digital tools within rural Mozambique specifically. This raises a clear concern that shadows digital development interventions more broadly: does digitalisation offer a genuine opportunity for transformation in participatory extension models, or does it risk becoming a form of technological utopianism that, given the cost of access and the skills required to use it meaningfully, ends up reproducing and deepening the very inequalities it aims to address (Toyama, 2015; van Dijk, 2005)?

This thesis considers these dynamics when examining how the digitalisation of FFS is communicated to participants in Nampula, with a particular focus on the introduction of smartphones by FAO. The aim is to understand how the introduction of smartphones is communicated and negotiated within the community, to explore how digitalisation is reshaping communication practices, and to assess the extent to which these processes reflect the participatory principles that FFS was originally built on.

The central research question guiding this study is: How do farmers interpret and negotiate the introduction of smartphones within the traditionally face-to-face learning environment of FFS

[1] Nampula, Zambézia and Manica are provinces in northern and central Mozambique. See Figure 1/ Annex 1 for a map locating these provinces. See Annex 2 for a map locating the districts of Nampula.

in Nampula? Two sub-questions support this analysis. The first asks to what extent does concentrating smartphone access among facilitators reinforce existing patterns of status within the community. The second examines what social and infrastructural barriers shape whether this digitalisation effort can translate into meaningful change or remain largely symbolic. Together, these questions allow for an assessment of how FFS digitalisation is experienced within the community, and whether it holds the potential for sustainable transformation.

To address these questions, the thesis draws on an interdisciplinary theoretical framework combining Heeks' (2002) design-actuality gap model, socio-technical perspectives from Information Systems (IS)[2], and van Dijk's (2005) four-level model of digital access, situating the case within broader debates on ICT4D and participatory communication. Empirically, the study draws on qualitative data collected during three weeks of fieldwork in Nampula province, that combined semi-structured interviews, focus groups and participant observation.

The thesis proceeds by first situating the case within Mozambique's agricultural and digital development context, before reviewing the relevant literature, outlining the theoretical framework and methodology in detail, and presenting the analysis and conclusions drawn from the fieldwork.

2. Background

2.1. Mozambique's agricultural and digital context

Mozambique's current agricultural landscape cannot be understood without its recent political history. Following independence from Portugal in 1975[3], the Front for the Liberation of Mozambique (FRELIMO) [4] government adopted a socialist model of agricultural collectivization, nationalizing land and organizing rural production into communal villages and state farms (Hall & Young, 1997; Pitcher, 2002). This approach generated considerable

[2] Information Systems (IS) is an academic discipline concerned with the study of how people, organisations and technology interact to collect, process, store and communicate information, with particular attention to the social and organisational context in which technology is designed, implemented and used (Avgerou, 2008).
[3] Mozambique was under Portuguese colonial rule from the late fifteenth century until independence in 1975. Portuguese colonialism was marked by resource extraction, forced labour and political domination.
[4] FRELIMO (Frente de Libertação de Moçambique) is Mozambique's dominant political party and has governed the country continuously since independence in 1975. Founded in 1962 as a national liberation movement against Portuguese colonial rule, FRELIMO's candidate in the 2024 presidential election, Daniel Chapo, took office as president in January 2025.

resistance among rural communities, including in Nampula province itself (Geffray, 1990; Bowen, 2000). A protracted civil war, lasting from 1977 until the 1992 Rome General Peace Accords[5], devastated rural infrastructure and livelihoods. From the late 1980s onward, and particularly following the end of the conflict, Mozambique underwent a significant neoliberal transformation characterized by market liberalization, privatization and an increasing reliance on private investment and donor-funded agricultural development programmes (Pitcher, 2002, pp. 3-5). These reforms laid the groundwork for the market-oriented and internationally funded agricultural development model that continues to shape agricultural interventions in the country today.

With approximately two-thirds of the country's 35.6 million inhabitants living in rural areas and a gross national income per capita of only USD 628.6 in 2025 (FAO, 2025b, p. 1; World Bank, 2026b, p. 1), Mozambique remains a low-income country whose economic stability and food security are closely tied to the performance of its agricultural sector, which employs approximately 70 per cent of the workforce while contributing only 27 percent of GDP (FAO, 2025b, p. 1). Family farming remains the dominant production system in Mozambique, with small and medium-scale farms accounting for 99.9 per cent of the country's approximately 5.2 million agricultural units (INE, 2026). Agricultural production is predominantly rainfed making the sector highly vulnerable to climatic shocks and weather variability (FAO, 2025b, p. 1).

[5] The Rome General Peace Accords was a peace treaty signed between the government of Mozambique and RENAMO, ending the Mozambican Civil War on October 4, 1992.

Figure 1. Map of Mozambique showing provincial boundaries. Source: Government of Mozambique (2010).



Mozambique's geographical position exposes it to recurrent extreme weather events, including cyclones, floods and droughts. Over the past two decades, the frequency and intensity of these events have increased, with severe consequences for agricultural production and rural livelihoods (Manuel et al., 2021). It is against this climatic vulnerability and persistent rural poverty that agricultural development has long been a priority for both the Mozambican government and international development partners. FAO has been present in the country since 1979, supporting the Government in addressing challenges across the agriculture, livestock, forestry, fisheries and aquaculture sectors. (FAO, n.d.-c). The FFS approach has become one of FAO's most influential interventions in Mozambique, offering a hands-on community-driven learning model that strengthens farmers' agricultural knowledge and decision-making capacities (FAO, 2016). More recently, *PROMOVE Agribiz*[6], a FAO programme financed by the European Union (EU), shows a shift towards a more integrated rural development strategy.

[6] *PROMOVE Agribiz* project webpage: https://www.fao.org/europeanunion/projects/projects-detail/promove-agribiz-gcp-moz-127-ec/en

The programme supports the development of small-scale sustainable commercial agriculture through agricultural training, like FFS and Farmer Business Schools (FBS)[7], value-chain support and social empowerment, particularly for women and youth (European Commission, 2024). Despite these efforts, structural constraints such as poor infrastructure and limited access to finance continue to shape development outcomes in rural Mozambique (FAO, 2025b, p. 1). These are constraints that digitalisation has increasingly been promised to address.

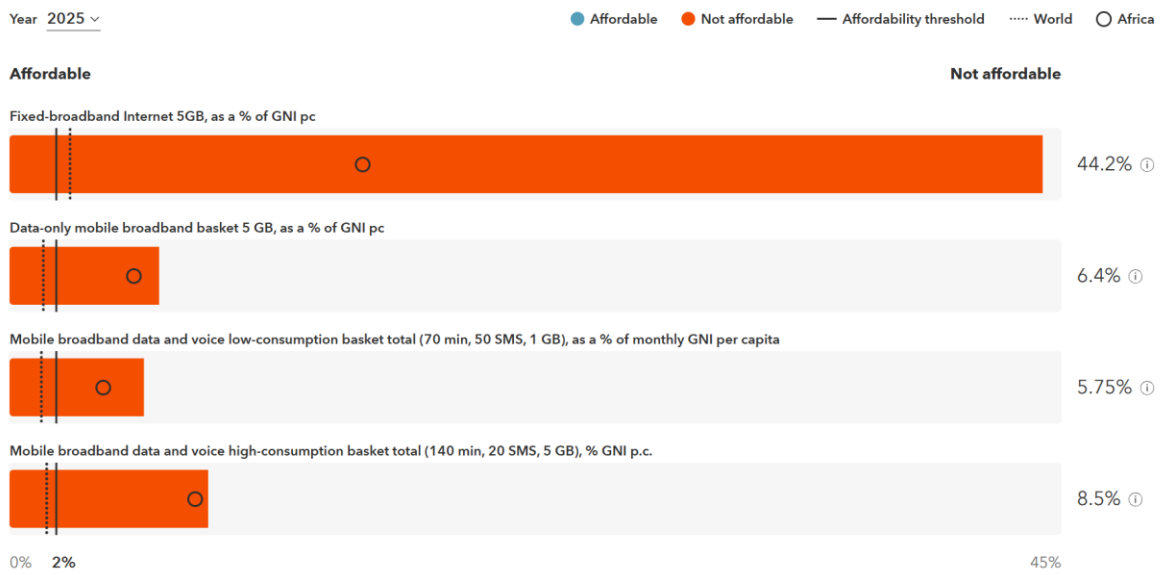
Despite ongoing investments in agricultural development projects, many smallholder farmers in Mozambique continue to face barriers that limit opportunities to move beyond subsistence farming, notably limited access to knowledge, skills and information (Adam & Quinhentos, 2018). Digitalisation has therefore emerged as a strategic priority to close this information gap and increase the reach and effectiveness of development projects. Digital tools can expand farmers' access to technical advice and peer networks (Khalid et al., 2025, p. 2). It is within this context that FAO, together with the Government of Mozambique, has pursued digitalisation initiatives through *PROMOVE Agribiz* (FAO, 2025a).

Digital readiness[8] remains a significant challenge for rural communities that shapes the adaptation and impact of ICT4D initiatives. In Mozambique, only around 20 per cent of the population uses internet (DataReportal, 2025), with connectivity particularly low in rural areas where access to electricity, mobile network coverage and device ownership remains limited (Torgusson & Pswarayi-Riddihough, 2022). While mobile phone adoption expanded rapidly in rural Mozambique between 2012 and 2016 following Movitel's[9] entry into the market, with the company's extensive investment in rural network infrastructure extending mobile services to previously underserved areas and increasing competition that lowered communication costs (Gillwald et al., 2019, p. 1), important barriers to digital inclusion persist. Access to electricity remains limited, with only 36.9 per cent of the population connected in 2024 (World Bank, 2024a), while the cost of a 5 GB mobile broadband basket represented 6.4 per cent of monthly Gross National Income (GNI) per capita in 2025, more than three times the Broadband

[7] Farmer Business School (FBS) is a training methodology that equips farmers with basic business and market-oriented skills, enabling them to manage farming as an enterprise through improved planning, financial management, and marketing decisions.
[8] Digital readiness refers to the skills, resources, and conditions, including digital literacy, affordability, infrastructure, and social context, that determine whether individuals or communities can meaningfully access and use digital technologies.
[9] Movitel is a Mozambican telecommunications operator launched in 2012 as a joint venture between Viettel (Vietnam) and SPI (Mozambique). It became a major driver of mobile expansion in rural areas by investing heavily in network infrastructure.

Commission's[10] 2 per cent affordability target and above both the African regional average of 5.32 per cent and the global median of 1.38 per cent (ITU, 2025).

Figure 2. Mozambique's 2025 ICT prices. Source: International Telecommunication Union (ITU) (ITU, 2025)



These national figures are also reflected among FFS participants. The unpublished FAO report, that informs this research, offers a detailed picture of digital readiness among FFS participants in Mozambique (FAO, 2026a). The assessment finds that over 40 per cent of FFS participants do not own any digital device, while fewer than 5 per cent use smartphones and fewer than 10 per cent have internet access. Women were almost three times more likely than men to not own a mobile phone, and even where household phones were available, women often faced male control over their use, alongside time poverty from combining domestic and agricultural responsibilities. Low literacy levels and language barriers present a fundamental obstacle to the use of text-based digital tools.

These structural limitations shape how digital tools are understood, trusted and used within agricultural settings. Digital readiness in rural Mozambique is not solely a matter of infrastructure, but also of local communication practices shaped by power dynamics and tradition. This suggests that the successful introduction of technologies such as smartphones requires approaches that extend beyond the provision of devices and connectivity, incorporating context-sensitive communication and capacity-building strategies, so that

[10] The Broadband Commission for Sustainable Development is a high-level international advocacy body that was established jointly by the International Telecommunication Union (ITU) and UNESCO in 2010.

communities can learn to use these tools effectively and integrate them sustainably into their daily lives.

2.2. FAO and ComDev

Understanding FAO's engagement with Communication for Development (ComDev) today requires looking back at the evolution of the discipline within the organisation. Development Support Communication (DSC) first emerged as a distinct field within the United Nations (UN) system at the end of the 1960s, with the first DSC service established in Bangkok in 1967, under the leadership of Erskine Childers and supported by United Nations Development Programme (UNDP) and United Nations Children's Fund (UNICEF) (Balit, 2012, p. 107). Following advocacy from UNDP official Childers, UNDP circulated a request to all UN agencies to include communication components in their development projects, leading to the establishment of DSC units across the UN system in 1969 (Balit & Acunzo, 2020, p. 136). FAO's own DSC Branch was established that same year under the leadership of Colin Fraser, with the objective of improving project planning, participation and capacity development. Although the Branch used the media resources from the Public Information Division, it did not deal with public relations or corporate communications, it was entirely field-oriented (Balit, 2012, p. 107). At the time, communication was largely understood as the dissemination of technical information to rural populations, and the unit struggled to gain institutional recognition within an organisation predominantly focused on agricultural sciences and technology transfer (Balit & Acunzo, 2020, pp. 137).

The year 2006 marked a pivotal moment when FAO, the World Bank and the Communication Initiative organized the First World Congress on Communication for Development in Rome, which resulted in the Rome Consensus. This document has as its key message that without communication there can be no development and defined ComDev as a dialogue-based process that promotes participation, builds trust, facilitates the sharing of knowledge and supports meaningful social change (FAO, 2007, pp. ix-x). Despite this progress, in the following years, FAO downgraded the discipline during a subsequent organisational reform process, the "Branch" was merged as a "Group" within the Extension, Education and Communication Service, limiting its impact and autonomy (Balit, 2012, pp. 110-112). The term "communication" increasingly became associated with public relations and corporate outreach. At the same time, the discipline has faced new challenges arising from globalisation, media

liberalisation, rapid socio-economic change and the emergence of new ICTs (Balit & Acunzo, 2020, p. 141).

Despite these institutional shifts, ComDev remains an active area of practice within FAO. FAO currently defines ComDev as follows:

> Communication for Development (ComDev) is a results-oriented and participatory communication approach that maximizes the impact and sustainability of development initiatives, facilitating informed decisionmaking, knowledge sharing and stakeholder engagement. ComDev is essential to advance inclusive rural communication services (RCS) by combining participatory processes with the use of media and digital solutions that can be appropriated by family farmers, rural communities and institutions. (FAO, n.d.-a).

This definition reflects FAO's continued emphasis on participation and dialogue as central elements of rural development. FAO's current ComDev work includes initiatives like Rural Communication Services, participatory video, rural radio and experience capitalization (FAO, n.d.-a).

Rural radio continues to play a central role as a communication channel for reaching farming communities. In Mozambique, this is exemplified by *Kianda*[11], a radio drama series developed by FAO and hosted through YenKasa Africa[12] to communicate agricultural and climate-adaptation knowledge to farming communities in Nampula and Zambézia (FAO, 2026b). According to FAO staff interviewed in Nampula, *Kianda* content is accessible on the smartphones distributed to the FFS facilitators (FAO staff, interview, March 2026). This link between radio content and smartphone access is explored further in the Analysis. The initiative illustrates how FAO combines traditional media such as rural radio with digital tools to expand access to agricultural knowledge (FAO, n.d.-a).

[11] Kianda episodes 2026: https://soundcloud.com/unfao/sets/kianda-radio-drama

[12] YenKasa Africa is a regional communication-for-development platform supported by FAO and partner organizations that connects community radios, media professionals, and rural development practitioners across sub-Saharan Africa (https://yenkasa.org/)

2.3. Farmer Field Schools

FFS are a group-based learning initiative in which farmers collectively test agricultural techniques, observe the outcomes and make decisions based on shared knowledge (FAO, 2016). This initiative replaces top-down teaching methods and transfer of technology approaches with a participatory learning process led by facilitators, who are farmers selected from within the community. This model supports sustainable implementation at the field level, as it builds on existing community capacities rather than external expertise (IFAD, 2022). Beyond agricultural knowledge, FFS sessions address related social issues, such as gender equity, making it a platform for community learning adjusted to farmers' needs rather than a purely technical training exercise (FAO technician, interview, February 2026).

> The FFS approach is an innovative, participatory and interactive learning approach that emphasizes problem solving and discovery based learning. FFS aims to build farmers' capacity to analyze their production systems, identify problems, test possible solutions, and eventually encourage the participants to adopt the practices most suitable to their farming systems. (FAO, 2011, p. 25)

The FFS approach is today widely implemented across the world. According to United Nations Department of Economic and Social Affairs (UN DESA), FFS programmes are currently active in more than 90 countries and empower an estimated 400,000 to 1 million farmers each year. The typical seasonal cost of these programmes ranges from USD 800 to USD 1,550, or roughly USD 40–80 per member (UN DESA, n.d.). The approach was first introduced by FAO in 1989 in Indonesia, as part of a programme promoting Integrated Pest Management (IPM) among rice farmers (FAO, 2016, p. 18). Since its introduction, the FFS approach has been widely recognised and adopted internationally. Despite this success, the implementation of FFS has faced several challenges. The quality and duration of facilitator training have been considered insufficient, and while monitoring and evaluation systems existed in many programmes, the human and technical capacities required to manage and analyse data were often lacking (van den Berg et al., 2021).

In Mozambique, FFS was introduced in 2001 through FAO's *Special Food Security Project*, which established 124 FFS groups between 2001 and 2005 (World Bank, 2024b, p. 4). Since then, the approach has expanded considerably, with FFS now implemented at scale across multiple districts. Between 2016 and 2021, FAO implemented the project *Strengthening Capacities of Agricultural Producers to Cope with Climate Change through the Farmer Field*

School Approach, which placed particular emphasis on climate resilience (FAO, 2023). Under *PROMOVE Agribiz*, a USD 53.4 million programme financed by the European Development Fund (EDF) and implemented by FAO in partnership with the Government of Mozambique between 2019 and 2025, the FFS model has aimed to strengthen market access and value-chain development in central and northern Mozambique (FAO, n.d.-d). The most recent initiative related to FFS in the country is the project *"E-Empowerment" of Small-Scale Producers: Leveraging Digital Tools Sustainably in Farmer Field Schools (FFS)*, launched in April 2025 (FAO, n.d.-b).

According to a technician working for FAO in Ribáuè, the implementation of FFS in Nampula's communities follows a structured process (FAO technician, interview, February 2026). As described in the interview, FAO and the Government of Mozambique begin by selecting communities where agriculture is the main production activity. FAO staff meets with local leaders to introduce the initiative and identify facilitators, typically an experienced farmer from within the community who can read and write in Portuguese. The facilitators receive training from FAO technicians and once trained, he or she forms a voluntary self-selected group of around 20–30 farmers. The group jointly selects a crop or production system to test and establishes a shared field divided into two sections: one to follow farmers' existing practices, and another one to test improved techniques such as new seeds, pesticides or irrigation methods. Members meet regularly to observe changes, compare results and make decisions collectively. Beyond agricultural experimentation, FFS sessions also address broader community issues, including nutrition, health and gender dynamics. Participants frequently share concerns on other daily life matters with the technician, who invites relevant specialists to speak on these topics. Once the testing phase is completed, the group invites the community to share the knowledge they have acquired. According to the FAO technician interviewed, FAO staff follow up on the results to encourage farmers to adopt the new practices on their own farms.

The FFS approach has been successful for many years across different countries because it can be adapted to the reality of each community. Many aspects of the process are decided collectively by participants, including which crops to work on, which techniques to test, how often to meet and other organisational details (FAO, 2016). Beyond agricultural learning, FFS promotes positive personal transformation, more equitable gender roles and household economic development (Duveskog et al., 2011). FFS also requires relatively limited financial investment, and, with an adequate exit strategy, it can be sustained by communities beyond the organization's active involvement. At its core, FFS is designed to be carried out and maintained

by participants themselves, placing decision-making power directly in farmers' hands (van den Berg et al., 2021). Table 1 presents the distribution of FFS groups, facilitators, and farmers across five districts in Nampula in 2025.

Table 1: FFS number of participants by District, Nampula (2025). Source: Data provided by FAO staff working with FFS in Nampula (personal communication, May 2026).

District	FFS number	Facilitators	Active members	Graduated members
Angoche	73	73	578	1,336
Mogovolas	66	66	489	1,275
Meconta	41	41	286	778
Ribáuè	86	86	836	1,303
Malema	61	61	458	1,082

Note. Each FFS group is led by a single facilitator, which is why the FFS Number and Facilitators columns are identical.

2.4. Digitalisation of FFS

FAO identifies farmers' access to information systems as an area that needs to be better explored, particularly in contexts where internet connectivity exists but digital skills remain limited. (FAO, 2023, p. 28). As discussed in FAO's webinar series *Leveraging Digital Innovations in FFS*, digital tools offer a way to enhance learning, strengthen collaboration and improve farmers' access to skills and markets. However, FAO also cautions that poorly designed digital strategies can worsen existing inequalities and weaken the quality of training. Understanding local digital access, skills and patterns of use is therefore key to effective and inclusive digital integration (FAO, 2025e)

Two FAO initiatives are currently driving the digitalisation of FFS in Mozambique. *PROMOVE Agribiz* includes a strong digital component. In 2024, the programme equipped FFS facilitators with smartphones, donated by Oukitel [13], to support data collection, communication with FAO staff and technicians, access to market and climate information, and digital financial services (FAO staff and technicians, interviews, February and March 2026). In addition to the device, facilitators received a protective case, a screen protector and a charger. They were required to sign a responsibility agreement outlining conditions for the use and maintenance of the devices. Fieldwork observations and focus group discussions with

[13] Oukitel is a Chinese technology brand best known for manufacturing rugged smartphones and tablets, and portable power stations. Website: https://oukitel.store/

facilitators who received the devices confirmed that, two years later, the smartphones continue to be used in day-to-day FFS activities as well as for personal purposes.

The second initiative supporting the digitalisation of FFS in Mozambique is the Flexible Voluntary Contribution [14] subprogramme under the FAO's Global FFS Platform, *"E-empowerment" of Small-Scale Producers: Leveraging Digital Tools Sustainably in Farmer Field Schools (FFS)*, implemented between February 2024 and October 2026. According to the project webpage, the initiative has a budget of USD 800,000 and operates in Mozambique, Malawi and the Syrian Arab Republic. It is a subprogramme designed to develop global digital products and skills that complement ongoing FFS activities. It is being pilot tested within existing projects, including *Building Local Resilience in Syria (BLRS)* and *PROMOVE Agribiz* in Mozambique, before being scaled further (FAO, n.d.-b). The objective of the *"E-empowerment"* project is to increase the digital engagement of FFS participants by exploring how digital tools can enhance the impact of FFS interventions.

As an initial step, FAO conducted a digital readiness assessment in the project's two pilot countries, Mozambique and Syria, to identify local needs and norms affecting digital adoption among FFS participants. According to a FAO staff member who is leading this initiative, the project has experienced delays[15], and current efforts have focused on finalising the *Digital Readiness Assessment for FFS Participants and Facilitators in Mozambique*, based on data collected in September 2025 (FAO staff, interview, February 2026). The report aims to assess FFS participants' levels of digital access, skills and tool usage; identify communication patterns within FFS contexts; examine barriers and enabling factors shaping digital engagement; and generate recommendations for sustainable digital empowerment strategies (FAO, 2026a).

Access to this report was particularly valuable for this research, it provides recent and large-scale data on the local context. Its findings point to notably low levels of digital readiness among FFS participants in Mozambique, a pattern examined further in the Analysis chapter.

[14] FAO's Flexible Voluntary Contribution (FVC) is a pooled and multi-donor funding instrument that provides flexible resources to support innovative and strategic programmes aligned with FAO's Strategic Framework and Sustainable Development Goals.
[15] The reasons for the project delays were not disclosed during the interview.

3. Literature review

The field of ICT4D has evolved significantly over the past decades, yet it continues to struggle with a persistent tension between technological innovation and social realities. Early ICT4D initiatives were largely driven by a modernisation paradigm that framed technology as a catalyst for economic growth and development, with Rostow (1960) and Lerner (1958) laying the foundations of this modernisation theory. Within this perspective, digital tools were frequently promoted as universal solutions, often with insufficient attention to local contexts, socio-cultural dynamics and existing knowledge systems. Kleine and Unwin (2009) argue that ICT4D has given limited consideration to the lessons learned from earlier information and communication initiatives, underscoring the need for deeper engagement between ICT practitioners and development studies.

A central critique of the modernisation perspective comes from Heeks (2002), who argues that many ICT4D projects fail because they are designed on assumptions that do not match the realities of the environment in which they are implemented. His concept of the design-actuality gap captures the mismatch between the expectations embedded in a technological system and the conditions in the field, including infrastructure, institutional capacity, skills and cultural practices, with the likelihood of project failure increasing as this gap widens (Heeks, 2002). Although developed in early 2000s, this critique remains particularly relevant to rural development, where structural constraints, such as weak connectivity, low digital literacy and socio-economic inequalities, shape how technologies are interpreted and used. Despite these well-documented challenges, many ICT4D interventions continue, in practice, to adopt a technology-first approach that prioritises innovation over contextual adaptation, it assumes that the introduction of digital tools will generate development outcomes on its own. Toyama (2015) challenges this techno-centric logic, arguing that even in an era of "amazing technology," digital tools alone cannot drive social change. The impact of technology ultimately depends on human actions.

In 1997, Chambers introduced the question of “whose reality counts?”, challenging development interventions designed according to the assumptions of external professionals rather than the experiences and needs of local communities. Similarly, Heeks (2002) proposes that ICT systems are frequently designed according to models derived from high-income contexts and transferred to low-income settings without adequate adaptation, an uncritical technology transfer. This critique is echoed by other scholars in the field of IS, such as Avgerou

(2008) and Walsham (2012), who challenge the one-size-fits-all approach to ICT and emphasise that an information system is a social system that embeds technology. From this perspective, digital tools are socially constructed, and their effectiveness depends on how they are interpreted, appropriated and integrated into the everyday practices of their intended users, a process that is shaped by social, institutional and cultural contexts (Avgerou, 2008; Walsham, 2012).

In rural contexts, the mismatches between design technologies and local realities are particularly evident. Information and Communication Technologies for Agriculture (ICT4Ag) initiatives tend to assume that farmers will adopt digital tools once they are made available, often overlooking barriers such as practices, literacy, language, trust and gender dynamics (Aker, 2011; van Dijk, 2005). ICT4Ag has nonetheless gained prominence as a strategy to increase agricultural productivity by improving market access and strengthen climate resilience, closely linked to the broader agenda of Climate-Smart Agriculture (CSA), which aims to enhance resilience and climate adaptation in smallholder farming systems (Lipper et al., 2017). Within this framework, digital tools such as mobile phones and apps are increasingly promoted as solutions for smallholder farmers, supporting climate adaptation through improved access to information, weather data and decision-making tools (World Bank, 2017, pp. 3-5). This broader trend is particularly evident in the rapid growth of mobile money as a rural development tool in northern Mozambique. Carvalho et al. (2020) found that the expansion of mobile money agents substantially increased uptake of digital financial services in communities that previously had little or no access to formal banking, positioning mobile money as a mechanism to expand economic opportunities and financial inclusion.

However, the effectiveness of digital tools in agricultural settings remains contested, with many ICT4Ag initiatives replicating the techno-centric biases identified in broader ICT4D interventions. Baumüller (2018) argues that digital agricultural projects often fail to consider farmers' actual information needs and communication practices, which results in low adoption rates. Empirical evidence from other countries illustrates this unevenness in practice. In Nigeria, Ogbeide and Ele (2015) found that smallholder farmers spent more on their mobile phone bill seeking market information than on any other agricultural use, while also reporting little benefit from access to weather information. In Tanzania, Churi et al. (2012) found that although mobile phones were increasingly used to share agricultural information, broader ICT tools, particularly internet-based services, remained underutilized due to low literacy levels, high costs and limited-service provision in rural areas. A comprehensive evaluation of new technologies must

therefore go beyond initial adoption and short-term outcomes to consider their scalability under real-world conditions in rural areas (Kumar & Doan, 2025, p. 4). Limited digital literacy, high device costs and poor connectivity continue to pose greater obstacles than the lack of technology itself, as Toyama (2015) argues through his "Law of Amplification", technology does not create human or institutional capacity, it amplifies existing capacities and inequalities. In rural Mozambique, where access to resources is unevenly distributed and shaped by gender and social hierarchies, these capabilities are often limited. As a result, the introduction of digital tools may have a limited impact or risk reinforcing existing inequalities.

ComDev offers a people-centred framework that relies on participation and dialogue. It emerged as a response to top-down development approaches, that advocates for communication processes that empower communities and support collective decision-making (Servaes, 2008). This participatory paradigm aligns closely with the principles of the FFS initiative, where farmers are positioned as active agents in the learning process rather than passive recipients of information. Similar principles can be found in Chambers et al.'s (1989) Farmer First approach, which challenges conventional top-down models of agricultural development by recognising farmers as knowledge holders and innovators rather than passive recipients of technologies and external knowledge. This approach emphasises the importance of building on local knowledge and experiences when designing and implementing development interventions.

The introduction of digital technologies into participatory initiatives such as FFS raises important concerns. ICTs can enhance communication and access to information, but if not carefully integrated, they risk undermining the very participation they are meant to support. In Mozambique, Celestino (2018) argues that mobile phones can complement rather than displace existing participatory communication structures: accessing community radio through mobile devices expands reach and strengthens civic engagement. The key is for interfaces to remain simple, affordable and aligned with local communication practices. This suggests that digital tools can support participatory approaches but only when they reinforce, rather than replace, the channels communities already trust.

A key concern that emerged from the literature is the risk of digital exclusion, particularly along gender and socio-economic lines. In LMICs, women are often disproportionately excluded from digital technologies due to structural barriers such as limited access to devices, time constraints and social norms (Hilbert, 2011; GSMA, 2020b, p. 43). This pattern has also

been documented in rural Mozambique. In a study conducted in two villages in Manica province, Freeman and Mubichi (2017) found that radio, television and mobile phone ownership was consistently lower among women than men. When it comes to mobile phones, women tended to rely on shared household devices, whereas men were more likely to own phones individually. These findings suggest that the introduction of digital tools may reinforce existing hierarchies, as inequalities in access, skills and usage tend to mirror broader patterns of social stratification (van Dijk, 2005).

In response to the limitations of techno-centric approaches, there has been a growing emphasis on human-centred design in ICT4D and ICT4Ag. Heeks (2008) develops this perspective further through the concept of "ICT4D 2.0". While its earlier phase, "ICT4D 1", focused primarily on infrastructure and basic access, this more recent phase shifts attention towards users, the creation of local content and economic sustainability. Mobile phones have emerged as the central tool of digital inclusion, replacing earlier models centred on telecentres and computers. In this sense, ICT4D 2.0 promotes "innovation at the margins" (Heeks, 2008, pp. 29-30), favouring solutions developed by and for communities in LMICs and moving away from reliance on external funding toward self-sustaining technological models. Similarly, Calzada (2023) introduces a participatory and capacity-building approach, Enabling Rural Innovation (ERI), which focuses on local agency and the inclusion of marginalised groups in innovation and learning processes. By strengthening social and entrepreneurial capacities, ERI reinforces the idea that sustainable technological innovation depends not only on access to tools or adequate infrastructure but on the active engagement of its users.

The digitalisation of FFS represents an attempt to integrate ICTs into a participatory model. The literature evidence suggests that the success of digital tools in development projects depends less on the availability of technology and more on a deep understanding of local realities carefully reflected in implementation strategies. This includes how technologies are communicated, how users are supported in acquiring the necessary skills, and how infrastructure is developed to allow these tools to become part of everyday practice.

4. Theoretical framework

This thesis is grounded in an interdisciplinary theoretical framework that combines perspectives from ICT4D, IS and ComDev. Together, these perspectives provide the analytical

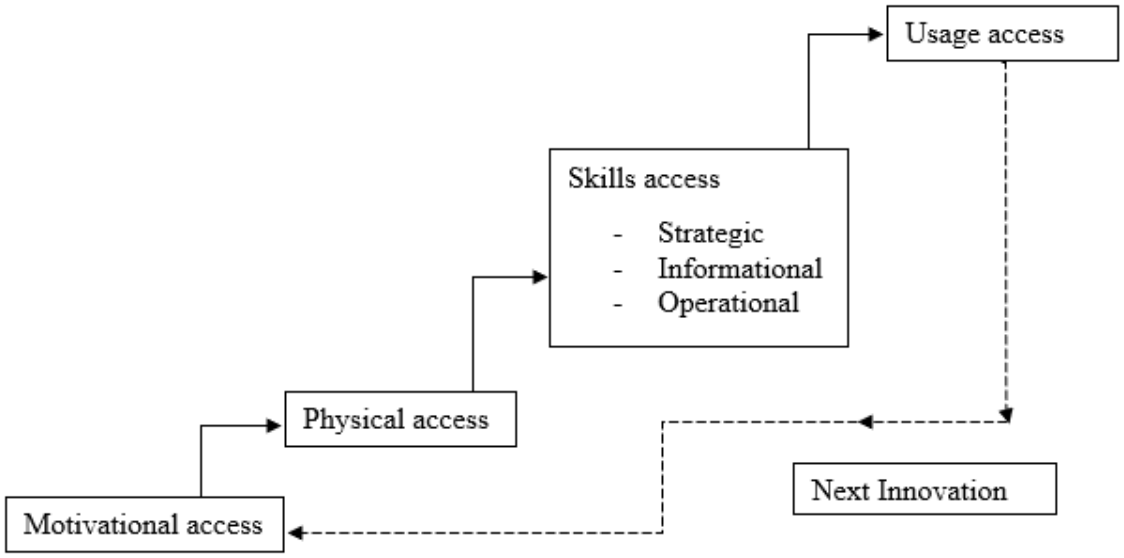
lens through which the digitalisation of FFS is studied. Rather than positioning digital technologies as neutral or purely technical tools, this study conceptualises them as socially embedded, shaped by contexts in which they are introduced and used, and capable of transforming communication models and community dynamics.

The central concept guiding this analysis is Heeks' theory of design-actuality gap. This framework frames the success or failure of ICT4D initiatives because of the alignment between the assumptions embedded in a technology's design and the realities of the local context, including users' skills, institutional capacity, infrastructure and socio-cultural practices. In this study, the concept is used as a diagnostic tool to examine how the strategy for introducing smartphones within FFS considered everyday practices and constraints of participants. It allows for a structural analysis of mismatches between the implementation strategy and farmers lived experiences as documented in fieldwork. While useful diagnostically, the design-actuality gap has been critiqued for assuming a relatively linear relationship between design intentions and local outcomes, which risks understating the extent to which local actors actively interpret and appropriate technologies rather than simply adopting or rejecting them as designed (Avgerou, 2008; Walsham, 2012). This limitation is addressed here by combining the design-actuality gap with a socio-technical perspective.

This thesis also draws on a socio-technical perspective from IS scholarship. Avgerou (2008) and Walsham (2012) argue that information systems are not merely technical but social systems in which technology and human practices are deeply intertwined. Following this perspective, this thesis aims to frame the effectiveness of digital tools as dependent on how they are interpreted and negotiated into everyday practices by its users. This approach shifts the analytical focus from the functionality of technology and toward the processes of meaning-making and adaptation of communication practices.

In addition, the thesis's framework incorporates digital inequality theory, particularly the work of van Dijk (2005), who conceptualises the digital divide as a multi-dimensional and dynamic process of social exclusion. His framework distinguishes four forms of access: motivational, physical, skills and usage. This perspective is especially relevant in rural agricultural contexts, where access to resources is uneven and shaped by social norms. By applying this theory, the thesis critically assesses the different forms of access experienced by FFS participants and how existing inequalities may be reproduced or reinforced through the introduction of digital tools.

Figure 3. Successive kinds of access to digital technologies (van Dijk, 2005, p. 22).



Throughout this study, ComDev serves as the foundational perspective guiding the analysis. This thesis examines how digital tools are introduced and communicated to participants, and how these processes shape their meaning and use. Particular attention is given to how the introduction of smartphones influences communication practices within FFS, including changes in information sharing, interaction and collective learning. In line with ComDev principles, this thesis explores the extent to which digital technologies contribute to participant empowerment by expanding access to information and communication channels, and whether this has the potential to improve livelihoods. Finally, consistent with ComDev's emphasis on bottom-up development, the thesis examines whether the introduction of digital tools is grounded in farmers' needs, the extent to which they were consulted, or whether it reflects an externally designed intervention with limited local input.

Collectively, these theoretical perspectives inform an integrated framework that conceptualises digitalisation as a socio-technical and communicative process shaped by context, power and participation. This thesis studies the processes through which digital tools are introduced, interpreted and negotiated in FFS, combining the design-actuality gap, a socio-technical systems approach, digital inequality theory and participatory communication. These perspectives support the analysis of the role of ICTs in participatory agricultural development initiatives and provide the basis for answering the research questions guiding this study.

5. Methodology

To analyse how farmers interpret and negotiate the introduction of smartphones within the traditionally face-to-face learning environment of FFS in Nampula, this thesis adopts a qualitative research design grounded in fieldwork. Since the research question is concerned with interpretation, negotiation and impact in communication practices, rather than with measuring rates of adoption, a qualitative approach is best suited to capturing participants lived experiences of digitalisation. Data collection was conducted over a period of one month and included three weeks of fieldwork in February and March in the province of Nampula, with the financial support of the Crafoord Foundation Travel Grant[16]. During this time, data collection combined semi-structured interviews, focus groups and participant observation. Given that only FFS facilitators received smartphones from FAO, data collection prioritised their perspectives, although focus groups were also conducted with FFS members who had not received a device. After this month of data collection, follow-up personal communication with FAO staff and technicians was carried out remotely to clarify emerging questions that arose during the analysis of the findings.

Two key informants played a crucial role in shaping the methodological approach and deepening the understanding of the institutional and social context surrounding FAO's presence in Mozambique. The first was a FAO technician from the province of Zambézia, who was contacted through LinkedIn during the early stages of the thesis. Drawing on his extensive experience in rural development and agriculture in the country, he provided valuable insights into the structure of FFS, the role of facilitators and the broader agricultural landscape in Mozambique. The second was a PhD researcher from Sapienza University[17] who is also studying the digitalisation of FFS in Mozambique. In September 2025, in collaboration with FAO, she travelled to Mozambique and conducted focus groups with FFS participants to produce a quantitative assessment of FFS groups' digital readiness. Her quantitative findings provided useful context for this study, allowing it to focus on the qualitative dimensions of smartphone use. Both key informants offered direct methodological support, advised on the design of the focus group guide, suggested appropriate ways to approach participants and shared relevant contacts.

[16] The Crafoord Foundation website: https://www.crafoord.se/en/
[17] Sapienza University website: https://uniroma1.web.uniroma1.it/en/

In total, five interviews were conducted with actors directly involved in the digitalisation of FFS: the *"E-empowerment"* project coordinator, the officer overseeing FAO projects in Nampula, the specialist responsible for FFS implementation in the province, and two FAO technicians allocated to the districts of Ribáuè and Meconta. In addition to these actors directly involved in the FFS digitalisation project, two further interviews were conducted with agricultural professionals outside the FAO structure, to provide broader context on the agricultural sector in Nampula; these are discussed later in this section.

In total, five focus groups were conducted during March 2026 with between 3 and 12 participants each (see Table 2). Two were held in Mogovolas and included a facilitator together with members of their FFS group. The remaining three were held in Ribáuè and included facilitators only. Ribáuè's focus groups took place during a FAO training session that had brought together facilitators from across the district, which was an opportunity to conduct the focus groups that would otherwise have required separate visits to each community and, therefore, would have not been possible logistically.

Table 2: Composition of focus groups by district, gender and participant type

Focus group	District	Composition	Participants	Women	Men
FG 1	Mogovolas	Woman facilitator + members	9	7	2
FG 2	Mogovolas	Man facilitator + members	12	8	4
FG 3	Ribáuè	Facilitators only	3	3	0
FG 4	Ribáuè	Facilitators only	8	0	8
FG 5	Ribáuè	Facilitators only	9	1	8

Initially, the plan was to equally involve both farmers and facilitators as FFS participants, and to organise focus groups by gender, on the assumption that participants would feel more comfortable speaking openly among others of the same gender, with an additional group reserved for young facilitators under the age of 35. Several of these plans changed once in the field. After consulting FAO's report on the digital readiness of FFS participants, it became clear that most farmers do not own smartphones (FAO, 2026a). And within the *PROMOVE Agribiz* project, FAO had provided smartphones exclusively to facilitators. For this reason, the study prioritised facilitators' perspectives, although two focus groups were still conducted with facilitators and their respective FFS groups, which include farmers. The plan to hold a dedicated focus group with young facilitators was also abandoned after discussion with the

specialist responsible for FFS in Nampula, who confirmed that very few young facilitators were active in the province, and that none were available in the districts where fieldwork took place.

While the original intention was to separate groups by gender, the final composition was shaped less by gender and more by proximity. Facilitators and their FFS members already knew one another well, which appeared to make participants more comfortable. In the facilitator-only groups, composition was largely determined by who chose to sit together, which resulted in groups formed around existing friendships rather than gender alone. Rather than viewing these changes from the original research design as deviations, they are better understood as a reflection of the realities of fieldwork, where the ability to adapt and find workable solutions in the moment is itself part of the methodological process.

For the focus groups with facilitators and FFS members, a translator was required, as most participants spoke only the local language, Macua. Translation was provided by an FAO technician rather than a professional translator, which may have introduced some loss or simplification of meaning during the exchange, as well as an additional layer of FAO's presence within these sessions. The facilitator-only groups did not need translation, since FAO requires facilitators to be proficient in Portuguese. The absence of a translator in these groups allowed for more direct interaction between me and participants.

An FAO technician or staff member was present at every focus group. Beyond providing translation when needed, their presence offered logistical support and a familiar face that likely helped put participants at ease. At the same time, FAO's presence carried risks. Participants may have associated me with FAO, potentially shaping their responses to what they perceived FAO would like to hear. The authority of FAO staff within the discussions may have influenced group dynamics and discouraged more critical reflections on the digitalisation process.

Participant observation was included to capture dynamics that might not surface directly in interviews or focus groups, such as the informal and practical use of the devices. Due to logistical and safety reasons, it was not possible to stay with the communities during the fieldwork. Instead, visits were organised to the communities to observe specific FFS activities. In Mogovolas, I spent time with an FFS group as they worked in the field alongside technicians[18]. In Ribáuè, I attended a training session for facilitators organised by FAO and

[18] Refer to Figure A3.1 in Annex 3 for a field photograph of Mogovolas FFS participants and technicians working together in the field.

delivered by the Japanese company ADM (Agro-Negócio para o Desenvolvimento de Moçambique Lda.) on tree production[19]. This session reflected a broader shift in FAO's approach as the *PROMOVE Agribiz* project nears its end and uses the FFS structure to connect farmers with future business opportunities.

As noted above, interviews were also conducted with two agricultural professionals outside the FFS structure to provide broader context on the agricultural sector in Nampula: a professor from the Faculty of Food and Agricultural Sciences at Rovuma University[20][21], and the director of the Instituto Politécnico Familiar Rural de Natete (IPFRN), a professional agricultural school linked to a Christian organisation.

Beyond these formal interactions, fieldwork in Nampula generated numerous informal conversations with community members, many of whom were eager to discuss their own realities and challenges given how central agriculture is to daily life in the region. These exchanges informed my broader understanding of the difficulties facing farmers in Mozambique. This kind of contextual knowledge could not have been acquired without fieldwork.

All interview and focus group participants were informed of the purpose of the study and their right to withdraw at any time. Verbal consent was obtained before each session. To protect the identities of interviewees, individuals are referred to by role rather than by name throughout this thesis.

Interview and focus group data were analysed drawing on the concepts outlined in the theoretical framework. Fieldwork information was reviewed and organised around three broad analytical categories: the alignment or mismatch between the initiative design and the realities of participants' everyday practices (the design-actuality gap); the social and institutional dynamics shaping how digital tools were interpreted, consequently changing communication practices (a socio-technical perspective); and differences in access to smartphones among participants (digital inequality). This thematic structure guided both the preparation and the analysis of the data collected during fieldwork.

[19] Refer to Figure A3.2 and A3.3 in Annex 3 for field photographs of ADM training session.
[20] Rovuma University website: https://unirovuma.ac.mz/
[21] Refer to Figure A3.4 and A3.5 in Annex 3 for field photographs from the Faculty of Food and Agricultural Sciences of Rovuma University.

Several aspects of this study's design affect how far its findings can be generalised. Facilitators, rather than farmers, were the primary respondents. This study speaks most directly to how digitalisation is communicated to and experienced by facilitators, and only indirectly, how it is understood by other farmers. The focus groups were limited to two districts, Mogovolas and Ribáuè, out of five districts covered by the *PROMOVE Agribiz* programme in Nampula. Findings should therefore be read as indicative of dynamics within these specific communities rather than generalisable across the province or the programme.

5.1. Researcher positionality

At the time of fieldwork, I was completing an internship with the International Fund for Agricultural Development (IFAD). This UN agency works closely with FAO and implements FFS initiatives. This position provided facilitated access to documents, contacts and logistical support during the preparation of fieldwork. However, this internship was unrelated to the FFS digitalisation programme or to FAO's *PROMOVE Agribiz* programme. The knowledge produced by this study was developed exclusively for this thesis and was not intended for use by FAO or IFAD beyond the academic context of the research. For this reason, the internship was not disclosed during focus groups, where wanted to position myself solely as a student, as I believed this would encourage participants to speak more freely.

It was difficult to create distance from FAO in the field. Travel to all communities took place in FAO vehicles, accompanied by FAO staff[22], who were present during the focus groups, as discussed above. This visible association with FAO may have shaped participants' perception of me.

The physical setting of the focus groups also varied and may have introduced further, subtler dynamics of authority. In the first location, I was seated on a chair while participants sat on the floor, this may have reinforced a sense of hierarchy. In the second location, the focus groups were held in a classroom where both the participants and I sat on student desks, a more level arrangement. These contrasting seating arrangements demonstrate how easily unequal power dynamics can be unintentionally built into fieldwork.

[22] Refer to Figure A3.6 in Annex 3 for a photograph from the field showing FAO vehicle and field personnel who accompanied visits to the communities.

My identity inevitably shaped the research relationship. Given the complex colonial history between Portugal and Mozambique, being Portuguese, white, and a young woman may have positioned me, in the eyes of the participants, as connected to a broader history of European presence and authority in the region. Finally, the limited time available for fieldwork further shaped the research process. Relationships of trust had to be built quickly; discussions may not have reached the depth that a longer-term engagement with these communities might have allowed. These dimensions of positionality are worth acknowledging, as they inevitably shaped the data produced in this research.

6. Analysis

This chapter analyses the digitalisation of FFS drawing on information collected during focus groups and interviews conducted during the fieldwork in Nampula. It approaches the introduction of smartphones as a communicative process, in line with ComDev perspectives, and a socio-technical process, that conditions how the devices were explained to recipients, how facilitators and farmers negotiated them in their everyday practices, and how structural inequalities in access shaped these dynamics. The chapter is therefore organised into three sections. The first examines how the introduction of smartphones was communicated, including the training provided and the follow-up support. The second analyses how facilitators appropriated and repurposed the devices in practice, drawing on Heeks' design-actuality framework to identify divergences between intended and actual use. The third analyses the structural constraints that shape unequal access to and use of technology, drawing on van Dijk's four forms of access. These sections address the research questions guiding the study by assessing the effectiveness and inclusiveness of the communication strategies used in the digitalisation of FFS.

6.1. Communicating the introduction of smartphones

The way in which smartphones were introduced to facilitators shaped how the devices were understood and used. According to FAO staff, no specific communication strategy was developed to support the introduction of smartphones to facilitators (FAO staff, interviews, February and March 2026). The initiative relied on a single training session delivered when

participants received the phones[23], with no structured plan for how this technology would continue to be explained over time. The training delivered by FAO technicians covered basic device functions (FAO staff and technicians, interviews, February and March 2026). Facilitators were consulted about what they wanted to learn in the training session (FAO staff, personal communication, July 2026). However, for most facilitators this was their first time using a smartphone, so they had limited knowledge about the devices and may not have been in the position to identify which functions would be most useful for them. This suggests a tension with Chambers' (1997) question of "whose reality counts?", as the training approach assumed that facilitators could identify their own learning needs despite having little or no prior experience with smartphones.

Facilitators who could not attend that session received their phones later but did not receive the training (FFS facilitators, FG 4 and FG 5, March 2026), this means that access to the device and access to the knowledge needed to use it were not guaranteed together. In addition, no manuals or other reference materials were provided to support learning after the training session.

The initial training went from learning how to turn the device on and off, to learning how to use WhatsApp, KoboToolbox[24] and radio (FAO staff, interview, March 2026). However, it did not include any guidance on how to access weather or market information, nor on how to use the e-voucher system[25], even though these are functions often identified by FAO as a priority (FAO, 2025a). This is consistent with what facilitators reported when asked what they use their phones for: weather and market information, and e-voucher access were rarely mentioned, while calls, messages, WhatsApp and radio content, were consistently identified as the phones' primary uses (FFS facilitators, FGs 1-5, March 2026).

At the time of fieldwork, no other formal learning opportunities have been provided or were planned. District-level technicians are responsible for providing ongoing support (FAO staff, interview, March 2026). However, when asked who they turn to when they have a question about how to use the phone, most facilitators named the spouse, a family member, or a neighbour rather than the technician (FFS facilitators, FGs 1-5, March 2026). This situation

[23] Refer to Figure A4 in Annex 4 for a photography of the moment FFS facilitators in Zambézia Province received the smartphones from FAO.
[24] KoboToolbox is a software used to collect, analyse and manage data for surveys, monitoring, evaluation and research. Website: https://www.kobotoolbox.org/
[25] The e-voucher system refers to FAO's digital mechanism, through electronic vouchers, for distributing agricultural inputs to farmers.

can be understood as a design-actuality gap (Heeks, 2002), while technicians were formally assigned the role of providing ongoing support, facilitators primarily relied on existing informal networks within the household and community when seeking assistance with their smartphones.

Interviews with FAO staff suggest an underlying assumption that, once facilitators received the devices, they would independently explore the phones and gradually learn additional functions through use (FAO staff, interviews, February and March 2026). However, this was not observed. Facilitators, when encountering an unfamiliar application or feature that had not been explained during the training, would close it immediately, due to the fear that improper use might damage the device (FFS facilitators, FG 3 and FG 4, March 2026). As a result, facilitators use their phones, but largely within the boundaries of what they were originally taught, rather than exploring other functions that the phones can offer. This finding illustrates the significant influence that digital skills training has on the adoption of new technologies. Trainings should be developed with farmers and extension services to target dimensions that are relevant to their needs (Xu et al., 2026, p. 11). These findings are consistent with Toyama's (2015) argument that technology does not generate capacity by itself, rather it amplifies existing capacities and support structures. In the absence of further training, existing social relationships became the actual mechanism through which digital skills were reinforced. Many facilitators explicitly stated that the training they received was insufficient, and that additional sessions were what they needed most (FFS facilitators, FGs 1-5, March 2026).

6.2. Negotiating the smartphones in everyday practices

How facilitators use the devices in practice reflects how technology was interpreted and adapted to their needs. WhatsApp was consistently identified as the most used function in FFS activities (FFS facilitators, FGs 2-5, March 2026). Before the introduction of smartphones, farmers depended on visits from district-level technicians to solve problems, from complex issues such as crop disease or pest infestation to more routine questions about FFS management. Even minor decisions were often delayed until a technician could travel to the community, which slowed down both problem-solving and day-to-day activities. With smartphones, facilitators began using WhatsApp to ask questions and share images directly with technicians to request advice in real time. This reduced their dependence on technicians' physical availability (FFS facilitators and FAO staff, FGs 1-5 and interview, March 2026).

FAO assumed that providing facilitators with the devices would improve farmers' trust in them as credible sources of knowledge. This expectation was linked to a broader concern that farmers tend to place more trust in the technical expertise of FAO technicians and staff than in the knowledge of facilitators, who are themselves farmers (FAO staff, interview, February 2026). In practice, however, facilitators often used the phone to call a technician on speaker, allowing the technician to directly confirm or clarify the information the facilitator was providing (FAO technician, interview, February 2026). This reflects a design-actuality gap (Heeks, 2002), as the expectation that smartphones would strengthen facilitators' credibility contrasts with their actual use as communication intermediaries between farmers and technicians.

Alongside communication with technicians, some facilitators also mentioned using their smartphones to listen to radio, either music or FAO's farmer radio programme mentioned before, *Kianda*, while carrying out agricultural tasks (FFS facilitators, FG 1 and FG 2, March 2026). This function was covered during the initial training, and although only a limited number of facilitators mentioned it, it suggests that when a function is taught, it is more likely to be adopted as intended. Facilitators explained that listening to music or the radio programmes makes agricultural tasks more enjoyable while also serving as a source of informal learning, since the radio programme conveys practical knowledge through storytelling (FFS facilitator, FG 1, March 2026). Launched in 2026 by FAO, the radio series is set in the fictional village of Kianda and follows a group of characters navigating the everyday reality of smallholder farming in Mozambique. Through storytelling, humour and relatable scenarios, the programme conveys practical knowledge on topics such as CSA, soil conservation, natural fertilisers, crop rotation, pest management and agribusiness planning (FAO, 2026b).

This use of the smartphone illustrates an effective combination of "old" and "new" media. From a ComDev perspective, radio remains a particularly effective medium in rural contexts because it provides a platform for local voices, facilitates the dissemination of development-related information and reaches communities that are often underserved by other forms of media (Johnson & Rajadurai, 2020). The integration of such content into smartphones represents a hybrid approach that combines traditional and digital media and that can improve both reach and usability. Facilitators further noted that, unlike text-based content, audio content could be listened to by several people at once, extending its reach beyond the facilitator holding the phone to farmers gathered nearby. Moreover, since audio content requires minimal literacy, it is accessible to farmers with varying literacy levels (FFS facilitator, FG 1, March 2026). This finding highlights the importance of aligning technological interventions with users' existing

practices and preferences. This reinforces the need for context-sensitive and user-oriented design in ICT4Ag initiatives.

Another function that emerged from the focus groups was the use of photography to support FFS activities[26]. Facilitators described using the smartphone camera to document different FFS activities and the outcomes of agricultural testing practices, creating a visual record that allows farmers to monitor changes over time (FFS facilitators, FGs 1-5, March 2026). These images can then be shared and collectively analysed during FFS sessions. Moreover, visual documentation supports communication with technicians by allowing farmers to illustrate problems more clearly than through verbal descriptions alone (FFS facilitators, FGs 1-5, March 2026). Photography represents a function of the smartphone that extends, rather than departs from, the pedagogy FFS was designed on. The camera becomes a new tool of observation and comparison. This exemplifies how new technologies can be integrated into existing participatory practices with relatively little friction, as their use aligns closely with what participants already do, unlike functions that require learning new skills.

Beyond FFS-related activities, some facilitators also use their smartphones for personal communication and for economic purposes. Several participants reported using WhatsApp to communicate with clients, share images of their products and negotiate prices (FFS facilitators, FG 4 and FG 5, March 2026). One example, documented in this study, shows a facilitator using WhatsApp to promote his poultry business[27]. While this use goes beyond the initial purpose of the devices, it remains a clear example of facilitators agency over an externally introduced technology.

Another observation, notable for its absence rather than its presence, is that facilitators never mentioned any social media platform beyond WhatsApp. This is worth noting given the documented growth of social media use in Mozambique, where the number of users increased by approximately 15.6 per cent between early 2024 and early 2025, reaching 3.70 million active users in January 2025 (DataReportal, 2025). Despite this broader national trend, no facilitator mentioned engaging with other social media platforms. One possible explanation is that the initial training focused exclusively on WhatsApp, reinforcing a pattern observed throughout

[26] Refer to Figures in Annex 5 for examples of pictures that the FFS facilitators took and shared with the smartphones.
[27] Refer to Figure A6 in Annex 6 for a screenshot of a WhatsApp message from an FFS facilitator promoting his poultry business.

this analysis: facilitators' use of the devices closely mirrored what had been explicitly taught, with limited experimentation beyond the boundaries set by the training.

Taken together, these findings show that facilitators neither adopted the smartphones exactly as designed nor rejected FAO's intended uses. Instead, they actively negotiated how the devices fit within their existing practices. At the same time, the boundaries of this negotiation were not unlimited. Facilitators' use of the smartphones remained closely aligned with what had been demonstrated during the initial training session. This dual pattern, active appropriation within a narrow training-defined space, supports a socio-technical understanding of digitalisation. From this perspective, technology does not carry a fixed or predetermined function; rather, it is shaped by practices of those who use it (Avgerou, 2008; Walsham, 2012).

6.3. Structural constraints and unequal access

While the previous section analysed how facilitators actively negotiated the smartphones functions within their everyday practices, this section turns to the structural conditions that shaped and constrained that negotiation. Access to a device is not, by itself, sufficient to change communication patterns. It is necessary to understand the different forms of access that determine whether, and how, a technology can be meaningfully appropriated to change communication practices in a rural community. This section uses van Dijk's (Figure 3) four levels of digital access to break down the structural barriers identified during the fieldwork into analytically separate dimensions. Rather than treating "access" as a single, undifferentiated problem, this framework helps identify the specific ways in which motivations, resources, skills and use shape the possibilities for meaningful engagement with technology. Motivational access was not found to be a barrier in this study: facilitators consistently expressed interest in the smartphones and said they would want another one if their current device stopped working (FFS facilitators, FGs 1-5, March 2026). It is important to note from the outset that the facilitators at the centre of this study, because they received a device from FAO, constitute a comparatively privileged group within a much larger population of FFS participants, for whom access to digital tools is far more limited (FAO, 2026a). The same is applicable to the broader Mozambican context, where digital access remains unevenly distributed (DataReportal, 2025).

The most immediate structural constraint identified in this study is the material access: the physical availability, affordability and maintainability of the technology. The barrier

mentioned the most frequently was the limited access to electricity in facilitators' households, which directly affects their ability to keep devices charged. Without electricity at home, facilitators rely on neighbours who do have access, usually paying a small fee to charge their phones (FFS facilitators, FGs 3-5, March 2026). This introduces a recurring cost and inconvenience related to something as basic as keeping the device powered. Material access is further shaped by the scarcity of retail shops and repair services in rural areas, combined with the relatively high cost of devices and repairs. The average price of smartphones had reduced in the last years, with the influx of sub-$100 devices from Chinese brands (GSMA, 2020a, p. 19), however, for many rural households it is still a significant expense. Several facilitators described struggling to find a place to repair a damaged phone or purchase a replacement locally and noted that repair costs were often prohibitive. It is also a struggle for simpler necessities: one facilitator who lost is phone charger explained that replacing it would require taking a day to travel to the city (FFS facilitator, FG 5, March 2026). As a result, some reported avoiding taking the smartphone into the field during daily agricultural activities, opting to use instead a more durable and easily replaceable feature phone. All facilitators expressed a clear interest in getting a replacement if their current device stopped working; however, when asked whether they would purchase one independently, most said they would not, mentioning cost as the main barrier (FFS facilitators, FGs 1-5, March 2026). This pattern illustrates that simply possessing a smartphone does not guarantee sustainable access to it.

These material constraints, as significant as they are for facilitators, appear considerably minor when set against the wider population of FFS participants. As discussed before, FAO's Digital Readiness Assessment reported substantially lower levels of device ownership, smartphone use and internet access among FFS participants overall. Women face particular disadvantages resulting in lower ownership rates, limited control over household devices and time poverty due to combined domestic and agricultural responsibilities (FAO, 2026a). As mentioned before, the facilitators at the centre of this study occupy a more privileged position than the farmers they work alongside. This disparity is not incidental to the initiative's design. Facilitators were selected as smartphone recipients precisely because of their existing community standing and higher literacy. As a result, the digitalisation of FFS was structured from the outset around an uneven distribution of access within the community.

Another structural constraint involves access to skills, or the digital competencies to use the smartphone effectively. Language and literacy emerged as relevant barriers, with several facilitators expressing a preference for voice calls over text-based communication, as writing

messages was perceived as more demanding, particularly for those with lower levels of literacy (FFS facilitators, FGs 1-3, March 2026). Vision-related difficulties were also frequently mentioned, with several facilitators noting the need for glasses to comfortably read text on the smartphone screen. As discussed in the previous sections, facilitators' use of the devices remained closely tied to what had been taught during the single initial training session. Unfamiliar functions were typically avoided rather than explored, largely due to concerns that improper use might damage the phone (FFS facilitators, FG 4, March 2026). Understood through the lens of skills access, this is not simply a matter of individual reluctance, but a structural limitation consequence of a training model that developed a narrow set of skills without providing the ongoing support needed to develop broader digital competences over time.

The final dimension considered is usage access, which refers to the extent to which technology, once available and learnable, is integrated into meaningful everyday use. In van Dijk's model, usage access represents the cumulative outcome of earlier forms of access. Limitations in material access and digital skills therefore tend to restrict the range of possible uses rather than merely delay them. This interpretation aligns closely with the findings from the focus groups. Features such as weather forecasts, market information, e-voucher access and social media were not mentioned by facilitators, despite being available on the devices (FFS facilitators, FGs 1-5, March 2026). An additional point that highlights the importance of usage access is the variation observed among facilitators who own the same smartphone model. Even when provided with identical devices, facilitators did not necessarily use them in the same way or for the same purposes. Ownership of identical technology does not therefore guarantee similar forms of engagement. Instead, the ways individuals use digital technologies are influenced by their skills, perceptions, needs and everyday contexts.

Gender should not be understood as a separate barrier operating alongside motivational, material, skills and usage access, but rather as a cross-cutting factor that shapes how each of these dimensions is experienced. Women are significantly less likely than men to own a mobile phone, and even when a phone is available within a household, control over the device often remains with male family members (FAO, 2026a). These constraints are further reinforced by the time pressures associated with domestic and agricultural responsibilities, which reduce opportunities to develop digital skills.

Interestingly, women who had been provided with FAO smartphones reported that men were generally less inclined to control or appropriate these devices, as they were perceived as FAO property and work-related tools (FAO staff, interview, February 2026). The association of the smartphones with FAO appeared to confer a degree of legitimacy and respect, which reduced some of the constraints that women commonly face regarding control over mobile devices. Nevertheless, ensuring that the digitalisation of FFS does not reinforce existing gender inequalities requires addressing barriers that extend beyond access to and control over technology. In Mozambique, studies show that women's digital participation is shaped mainly by affordability, time poverty and social norms (GDIP, 2026, p. 5). As a result, even when access to a smartphone is secured, gendered disparities may persist in the ability to effectively sustain and benefit from its use over time.

The findings presented in this section demonstrate that structural constraints on access are not peripheral obstacles to an otherwise well-designed initiative. They are fundamental to understand how the digitalisation of FFS is unfolding in practice. Motivational, material, skills and usage access are closely interconnected (van Dijk, 2005). Limited access to electricity, connectivity and affordability restrict what facilitators can reliably do with their devices; limited training restricts what they know how to do; and these factors shape and restrict patterns of use, despite facilitators' clear willingness to engage with the technology. Gender further influences each of these dimensions, with men and women experiencing access differently within the same programme.

When considered alongside the findings of the previous sections, these constraints reveal that the design-actuality gap (Heeks, 2002) extends beyond shortcomings in training or communication strategies. The gap is also embedded in the broader material, economic and social realities of rural Nampula, conditions that existed prior to the initiative and were not addressed by it. As a result, the introduction of smartphones into FFS cannot be understood as a neutral technological intervention. Instead, it interacted with existing structures of inequality that shaped who could access, learn and benefit from the technology and to what extent.

7. Conclusion

This thesis aimed to examine how farmers interpret and negotiate the introduction of smartphones within the traditionally face-to-face participatory learning environment of FFS in

Nampula, Mozambique. Based on evidence gathered during three weeks of fieldwork, through focus groups, semi-structured interviews and participant observation, this thesis analysed the distribution of smartphones to FFS facilitators as a socio-technical and communicative intervention embedded in broader institutional and social contexts. To interpret the information collected, it employed an interdisciplinary framework that integrates Heeks' design-actuality gap, socio-technical perspectives from IS scholarship and van Dijk's four levels of access.

The findings reveal a consistent pattern across the three dimensions examined: communication, negotiation and structural access. Regarding communication, no dedicated strategy was developed to support facilitators in adopting and integrating the smartphones into their everyday communication practices. Facilitators received a single training session at the time of the distribution of the phones, with no subsequent formal learning opportunities during the following two years. The findings indicate that training significantly shaped subsequent patterns of smartphone use, with facilitators rarely reporting the use of functions beyond those introduced in the initial training. The distribution of smartphones should not be understood as a purely technical intervention, but as a social intervention that requires resources allocated to digital skills development.

Regarding negotiation, facilitators neither used the smartphones exactly as intended by FAO nor rejected them. Instead, they actively adapted the technology to fit their existing needs and communication practices. The introduction of smartphones was therefore not a predetermined process; it was a negotiating process where the meaning and value of the devices were continuously shaped by facilitators' own priorities and local contexts. This finding is consistent with the Farmer First perspective (Chambers et al.'s, 1989), the introduction of smartphones was a negotiating process where the meaning and value of the devices was continuously shaped by farmers' active role as knowledge holders and local contexts.

In terms of structural access, the findings show that facilitators who received smartphones from FAO occupy a relatively privileged position within the broader population of FFS participants, many of whom continue to face significant barriers to device ownership and usage. The study highlights that motivation alone is insufficient to ensure digital access. Access is shaped by a range of factors that determine whether individuals can acquire, maintain and effectively use technology over time. Among these factors, gender emerged as a particularly significant dimension that influences both access to devices and opportunities to develop digital skills.

These findings suggest that Heeks' (2002) concept of design-actuality gap can be observed in the smartphone distribution process, where the intervention design did not adequately reflect the broader material, economic and social conditions of rural Nampula. This also echoes Chambers' (1997) question of "whose reality counts?", as the assumptions embedded in the intervention did not always align with facilitators' needs. The challenges observed in the field cannot therefore be understood solely as implementation issues, they are rooted in the context in which the technology was introduced. This supports a broader argument, consistent with socio-technical perspectives in IS (Avgerou, 2008; Walsham, 2012), that the meaning and functionality of a technology cannot be defined by design alone. They are negotiated through everyday use and shaped by skills and resources of those who engage with them. Digital interventions, when implemented without addressing existing structural inequalities, risk reinforcing, rather than resolving, the very disparities they are intended to address.

Since FFS effectiveness is rooted in its participatory and farmer-led approach to learning that challenges the conventional top-down extension models, this study raises important questions regarding the consequences of limiting the access to new technologies to a relatively small group of facilitators. Although facilitators were selected because of their experience, literacy levels and respected position within their communities, the unequal distribution of smartphones may unintentionally create new hierarchical tensions within FFS groups. This suggests a potential tension between the participatory principles that ground the FFS approach and a model of digitalisation in which access to technological resources is unevenly distributed. Such a tension merits careful consideration in future efforts to integrate digital technologies into participatory agricultural development programmes.

This study should be considered in light of several limitations. First, it focuses primarily on facilitators rather than the broader population of FFS participants. Second, fieldwork was conducted in only two of the five districts in which *PROMOVE Agribiz* operates in Nampula. My position, as a foreign, Portuguese-speaking student affiliated at the time with an UN organisation and travelling with FAO support, may also have shaped how participants chose to present their experiences, despite efforts to minimise this influence. Finally, the relatively short duration of the fieldwork provided only a snapshot of participants' experiences at a particular moment in time. These limitations suggest that the findings should be interpreted as context-specific insights into the digitalisation of FFS within the communities studied, rather than as conclusions that can be readily generalised to all Nampula, Mozambique, or FFS programmes more broadly.

This thesis should be viewed as a starting point for further research rather than a definitive account of the digitalisation of FFS. Future research could include a wider range of districts and socio-economic contexts, as well as explore how smartphone use and digital skills evolve over a longer period. Comparative research across countries could also help distinguish findings specific to rural Nampula from broader patterns related to the integration of digital technologies into agricultural development programmes.

Ultimately, this thesis suggests that the digitalisation of FFS in Nampula represents neither a transformative shift in learning, communication and decision-making processes nor a simple failure. Instead, it occupies the more ambiguous space anticipated by the theoretical framework, where the potential of digital technologies to support participatory agricultural development remains evident, but is shaped by the communication strategies and structural conditions through which these technologies are introduced, negotiated and used.

Declaration of AI use

This thesis was produced independently with limited usage of Artificial Intelligence (AI). Microsoft Copilot, made available by Malmö University, was used to support language editing and grammar correction. AI was also used in an initial phase, for the preliminary identification of concepts and academic sources.

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Annexes

Annex 1. Map of Mozambique

Figure A1. Map of Mozambique with provincial boundaries, provinces capitals and other relevant cities.

Source: Government of Mozambique (2010).



Annex 2. Map of Nampula Province

Figure A2. Map of the districts of the province of Nampula, including the new districts created in 2013.

Source: Government of Mozambique, provided by FAO staff, personal communication (2026).



Annex 3. Fieldwork photographs

Figure A3.1. FFS facilitators and technicians marking the field for planting.

Source: Author's photograph (Sara Alexandra Sousa, 2026).



Figure A3.2. FFS facilitators attending a training session delivered by ADM and organised by FAO.

Source: Author's photograph (Sara Alexandra Sousa, 2026).



Figure A3.3. Example of learning materials used during the ADM training session attended during fieldwork.

Source: Author's photograph (Sara Alexandra Sousa, 2026).



Figure A3.4. Faculty of Food and Agricultural Sciences, Rovuma University, Nampula Province.

Source: Author's photograph (Sara Alexandra Sousa, 2026).



Figure A3.5. Classroom at the Faculty of Food and Agricultural Sciences, Rovuma University, during a workshop session.

Source: Author's photograph (Sara Alexandra Sousa, 2026).



Figure A3.6. FAO vehicle and field personnel during a visit to an FFS community in Nampula Province.



Annex 4. Distribution of smartphones to FFS facilitators

Figure A4. FFS facilitators in Gurúè district, Zambézia Province, receiving the smartphones distributed by FAO.

Source: FAO staff member, personal communication (June 2026).



Annex 5. Photographs taken by FFS facilitators and shared via WhatsApp

Photographs of FFS activities taken using FAO-provided smartphones by facilitators and shared with FAO technicians via WhatsApp.

Source: FFS facilitators, personal communication (March 2026)







Annex 6. Example of use of WhatsApp for small-scale business activities

Figure A6. Screenshot of a WhatsApp message sent by an FFS facilitator to promote a poultry business. The message reads: “Good morning come buy chickens with me they are already in the phase of commercialization.”

Source: FFS facilitator, personal communication (March 2026).

